

MSAC512 — Predictive Modelling of Residential House Prices in Zimbabwe

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Module: MSAC512 — Machine Learning

Assignment: Practical Assignment 1

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Overview

This repository contains the complete code, data and results for a multimodal automated valuation model (AVM) built on web-scraped property listings from www.property.co.zw. The model combines:

- **Structured features** (bedrooms, bathrooms, floor and stand size, suburb, area)
- **Hand-built text features** (18 condition/finish flags extracted via regex)
- **NLP text features** (TF-IDF + truncated SVD)
- **Visual features** (8 CLIP-based condition scores from listing photographs)

The pipeline spans six assignment sections:

Section	Focus	Deliverables
A	Data acquisition, cleaning, scraping ethics	Raw and cleaned CSVs, ethics memo
B	Multimodal feature engineering	Feature dictionary, verbatim phrases, visual pipeline
C	EDA and statistical diagnostics	Price distribution, suburb summary, Kruskal-Wallis, ANCOVA, VIF
D	Predictive model development	GLM benchmark + 3 ML models, spatial split, conformal intervals

E	Interpretability and fairness	Residuals, permutation importance, local explanations, ablation
F	Madokero vs Mabvazuva case study	Point estimates, intervals, extrapolation, memo

Repository structure

```

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├── README.md           This file
├── requirements.txt     Python dependencies
├── Untitled8.ipynb     Main Jupyter notebook (all analysis)
├── Data/
│   ├── harare_listings_CLEAN_FINAL.csv
│   ├── house_prices_greater_harare.csv
│   ├── SECTION_D_multimodal_cohort_150.csv
│   ├── SECTION_B_feature_dictionary.csv
│   ├── SECTION_D_FINAL_VALIDATION_COMPARISON.csv
│   ├── SECTION_D_FINAL_TEST_WITH_CONFORMAL_INTERVALS.csv
│   ├── SECTION_E_global_permutation_importance.csv
│   ├── SECTION_E_group_ablation.csv
│   ├── SECTION_E_RF_residuals.csv
│   ├── SECTION_E_residuals_by_suburb.csv
│   ├── SECTION_E_residuals_by_area.csv
│   ├── SECTION_E_error_by_price_band.csv
│   ├── SECTION_F_synthetic_point_predictions.csv
│   ├── SECTION_F_synthetic_prediction_intervals.csv
│   ├── SECTION_F_location_title_decomposition.csv
│   └── SECTION_F_density_extrapolation.csv
├── Figures/
│   ├── FIG_condition_price_by_suburb.png
│   ├── FIG_RF_residuals_vs_fitted.png
│   └── FIG_RF_residuals_vs_floor_size.png
├── Report/
│   └── MSAC512_Report.pdf    Final written report

```

How to run

1. Set up the environment

```
conda create -n msac512 python=3.14 -y
conda activate msac512
pip install -r requirements.txt
```

2. Launch the notebook

```
jupyter notebook Untitled8.ipynb
```

3. Run the notebook end-to-end

Kernel → Restart & Run All

The notebook will:

1. Load the raw scraped data (already included in the repository).
2. Rebuild all features, models and figures.
3. Write all SECTION_* output files to disk.

Important: the notebook does **not** re-run the scraper by default — that step took several hours and requires live network access to www.property.co.zw. The raw scraped dataset is included so that all downstream analysis is fully reproducible offline.

4. Re-run the scraper (optional)

The scraper code is included in Section A of the notebook. To re-run it:

- Ensure the target site is reachable.
- Observe the rate-limiting delay (1–3 s between requests).
- Expect the full scrape to take several hours for 500+ listings.

Reproducibility notes

- **Random seeds:** All stochastic steps use `random_state=42`, except the training/validation split which uses `random_state=123` to preserve independence from the test split.
- **Computational cost:** Full notebook execution takes approximately 20–30 minutes on CPU. The CLIP feature-extraction step (Section B.3) accounts for most of this time.
- **Model artifacts:** The notebook does not persist model binaries; models are re-fit from scratch each run. This keeps the repository lightweight and guarantees reproducibility.

Methodological highlights

- **Spatial split** — three-way suburb-grouped split (train / validation / test) with **zero suburb overlap**, avoiding the location leakage that would contaminate a naive random split.
- **Nested hyperparameter tuning** — 5-fold inner GroupKFold on training data; validation used only for model selection; test set held out entirely.
- **Conformal prediction intervals** — distribution-free 90% intervals via split-conformal calibration on the validation set.
- **Feature ablation** — quantitative decomposition of model performance into size / location / condition / text contributions.
- **Density diagnostics** — 5-NN distance in 100-D feature space used to formally assess extrapolation for the synthetic case-study properties.

Data ethics

- robots.txt was inspected and respected.
- A randomised delay of 1–3 s between requests was enforced.
- Only publicly listed search and detail pages were scraped.
- No commercial use is made of the data.
- Currency was restricted to USD to avoid exchange-rate contamination.

See Section A of the report for the full ethics memo.

Disclosure

Generative AI tools were used for (a) code assistance during notebook development, (b) improving the readability of written sections, and (c) generating first drafts of some discussion sections. All numerical results, tables and figures are the direct output of the author's own code and were validated against the notebook.

Contact

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